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Autonomous intersection management: graph-reinforcement learning method based on road information enhancement

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ABSTRACT

Although extensive efforts have been made in the field of Autonomous Intersection Management (AIM), the transition to a fully autonomous traffic environment will require considerable time. In mixed traffic, the behavioural uncertainty of human-driven vehicles (HVs) poses a major challenge to stable cooperation among autonomous vehicles. To address this issue, this paper proposes a dynamic cooperation framework for multiple autonomous vehicles based on integrating Graph Attention Networks (GAT) with Deep Reinforcement Learning (DRL). The framework employs graph-based representations to model interaction relationships within the scene and innovatively incorporates prior knowledge of intersection topology into the representation process, helping to mitigate agent inaction caused by redundant interaction information. Safety constraints are introduced to restrict excessive exploration of driving policies, thereby enhancing driving safety performance. The effectiveness of the proposed method is validated through efficiency and safety metrics, and the reasons for performance improvements are further analyzed via individual agent behaviour.

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1. Introduction

Intersections have long been recognised as critical bottlenecks in urban traffic systems. Traditional signal-based traffic control strategies often trade off traffic efficiency to ensure safety and orderly flow. In contrast, Autonomous Intersection Management (AIM) enabled by Connected and Autonomous Vehicles (CAVs) has emerged as a promising alternative, offering the potential to significantly enhance intersection throughput by leveraging the advanced perception, communication, and control capabilities of CAVs (Guillen-Perez and Cano 2021). As the penetration rate of CAVs increases, their influence on intersection management will become increasingly pivotal. However, a transitional period characterised by the prolonged coexistence of CAVs and human-driven vehicles (HVs) is anticipated, introducing substantial heterogeneity into intersection environments. Consequently, under such heterogeneous traffic environments, achieving AIM through the coordinated driving of CAVs constitutes an emerging challenge for ongoing research.

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Currently, common AIM strategies rely on centralized cooperation methods, such as rule-based strategies (Kamal et al. 2015; Nichting et al. 2020) and system-optimal strategies (D. Zhou et al. 2022). These strategies utilise global state information to coordinate lane-level or vehicle-level traffic permissions. Unlike platoon control under signalised management, the control commands of these strategies are often directly transmitted to specific vehicles through communication networks, providing more flexible spatiotemporal access rights. However, these fully centralised methods are challenging to apply in mixed traffic environments. Human-driven vehicles in mixed traffic are not subject to centralised control instructions, causing difficulties for CAVs to handle unpredictable behaviours under predefined centralised strategies (Wu, Wang, and Zhu 2025). In comparison, distributed control frameworks have proven highly effective in managing multidimensional complex inputs, enhancing system efficiency, and improving robustness. AVs are regarded as independent agents capable of making autonomous decisions based on local perception, thereby effectively addressing the uncertainty posed by HVs. Additionally, global coordination achieved through information sharing ensures efficient intersection operations (Li, Cao, and Li 2021; Han et al. 2022). In our research, we aim to develop a distributed automated intersection management framework utilising multi-agent reinforcement learning (MRL), a widely adopted approach enabled by advancements in deep learning to address decision-making problems in cooperative and competitive multi-agent environments. During the learning process, each agent continuously interacts with the dynamic environment and utilises reward-driven feedback to learn optimal behavioural strategy (Guillen-Perez and Cano 2020). Therefore, the key to agent decision-making is environment representation and driving objectives.

Perceiving and understanding complex environments is an important prerequisite for agent decision-making (Kalantari et al. 2016). In mixed traffic environments, the interactions and states of vehicles dynamically change over time. Each vehicle's decision influences the behaviour of surrounding vehicles, making accurate representation of dynamic interactions a significant challenge for cooperative control among multiple autonomous vehicles. However, previous interaction modelling methods rely on direct vehicle relationships, such as perception and communication links (Ha et al. 2025; Klimke, Völz, and Buchholz 2023), while overlooking the role of road topology in vehicle interactions. In intersection environments, lane attributes determine vehicle movement directions, and the topological relationships between lanes directly dictate which vehicles have intersecting paths. When interaction information is incomplete, it becomes difficult to provide the agent with specific and valuable input for the learning process (Zhang et al. 2024; Liang et al. 2020). This inevitably causes agents to waste attention on irrelevant node information, leading to confusion in environmental perception and hindering their ability to make reasonable decisions.

In a different vein, traditional reinforcement learning guides agent decision-making through predefined reward functions, which are primarily designed based on human prior knowledge or management preferences. In multi-agent systems, designing an optimal reward function that balances efficiency and safety objectives in traffic systems is highly challenging. Moreover, agents often struggle to strictly adhere to implicit safety constraints embedded within the reward function, particularly in complex environments such as mixed traffic flow. This limitation presents a critical challenge for AIM.

To address the above challenges, this paper introduces a Hybrid Graph-Safe Deep Reinforcement Learning method (HG-SDRL) – a graph-based reinforcement learning framework

designed to facilitate agent cooperation and enhance autonomous driving decisions in highly interactive scenarios. Our model constructs a scene hybrid interaction graph integrating road topology information to represent the interactions among heterogeneous vehicle. Leveraging an attention mechanism, the framework captures deep reasoning capabilities for vehicle cooperation and potential right-of-way relationships. To ensure safe action exploration, the unsignalised intersection decision-making problem is formulated as a Constrained Markov Decision Process (CMDP), where a safety cost incorporating road topology priors is introduced as a constraint. Furthermore, we adopt a Lagrangian multiplier-based policy optimisation approach to improve the objective function and guide the learning process. Finally, we conduct an overall evaluation of the system performance of the proposed method in scenarios with different rates of penetration. Due to the introduction of prior information and safety enhancements, our method outperforms baseline methods in terms of both efficiency and safety. Furthermore, the attention-based explicit representation provides detailed insights into the reasons for performance improvement in the decision-making process. The main contributions of this research are as follows:

- (1) To achieve efficient and safe autonomous intersection management in mixed traffic environments, this research proposes a novel multi-agent cooperative control framework based on Hybrid Graph-Safe Deep Reinforcement Learning (HG-SDRL).
- (2) In this paper, a dynamic graph model of the intersection is established to describe the dynamic interactions between vehicles. The introduction of the intersection's topological information reduces redundant information exchange and significantly enhances the learning capability of the DRL model.
- (3) This study adopts a Lagrangian multiplier-based policy optimisation approach to address the trade-off between system efficiency and safety, thereby improving the safety performance of the decision-making process.

2. Related work

2.1. Autonomous intersection management

AIM has received extensive research attention in recent years, benefiting from the advancement of autonomous vehicles. AIM was first proposed by Dresner and Stone (2008), who introduced a vehicle reservation mechanism to replace traditional traffic signals at intersections under the assumption of full communication. The reservation mechanism is based on a First-Come-First-Served (FCFS) rule. The FCFS strategy ensures conflict-free passage by pre-allocating the right-of-way based on the sequence of vehicle arrivals. Since then, rule-based approaches have been widely adopted. Carlini, Boyles, and Stone (2013) proposed an auction-based intersection management mechanism that allows drivers to bid for their passage order based on their individual travel values. Building on the FCFS strategy, Chai et al. (2018) introduced a more flexible space-time slot control strategy, which permits multiple non-conflicting vehicles to enter the intersection simultaneously and optimises the passage order by utilising the safety gaps between vehicles. However, these rule-constrained policies provide feasible rather than optimal solutions for AIM. To this end, some studies have focused on obtaining optimal solutions for AIM. Kamal et al. (2015)

developed a model predictive control (MPC)-based global coordination method for signal-free intersections, aiming to enhance driving safety and stability by minimising collision risks and promoting smooth vehicle speeds. However, due to limitations in model solvability and scenario scale, their study did not address intersection efficiency. Levin and Rey (2017) proposed an autonomous intersection management framework by combining a mixed-integer programming model with a rolling optimisation window, optimising intersection throughput in fully autonomous environments by allocating occupancy of conflict points. Deng et al. (2023) treated vehicle platoons as decision variables and applied constraint programming to solve vehicle merging, platoon formation, and queue passing sequences. Compared to vehicle-level coordination methods, their approach demonstrated higher traffic efficiency and fairness. However, system-optimal strategies rely heavily on mathematical modelling, and as the number of agents and constraints in mixed traffic systems increase, solving the model becomes increasingly challenging.

To address the limitations of traditional methods, recent studies have adopted multi-agent reinforcement learning (MARL) algorithms to enhance the adaptability of CAV control strategies in dynamic environments. Wu, Chen, and Zhu (2019), as one of the early works applying MARL to AIM, decomposed agent states into independent and joint components based on intersection grid occupancy and designed both independent and joint action policies to control CAVs, aiming to reduce unnecessary coordination overhead. Guillen-Perez and Cano (2022c) proposed an end-to-end MARL-based AIM approach (ARAIM) that recursively integrates the state information of an arbitrary number of CAVs using LSTM and makes decisions using the TD3 algorithm and curriculum learning strategy. Due to extensive information fusion and robust training methods, their model demonstrated strong cooperative control performance across various traffic conditions. Building upon the aforementioned method, Guillen-Perez and Cano (2022b) enhanced the learning efficiency of their method through imitation learning, and further considered the impact of communication delays on multi-agent decision-making and extended their framework to larger-scale intersection networks (Guillen-Perez and Cano 2022a). Although these studies have shown promising progress in automated intersection management using MARL, they still lack adaptation to mixed traffic environments, and their ability to handle the uncertainty introduced by human-driven vehicles remains unclear.

2.2. Interaction of multiple driving agents

The modelling of interaction relationships aims to accurately capture the dynamic interactions between agents and their surrounding environment, thereby enhancing the utility of environmental state information. Traditional approaches that use the entire sequence of features as input fail to reflect these interaction relationships and are prone to the influence of redundant information (Guillen-Perez and Cano 2020; Liu et al. 2024a; Zhao et al. 2023; 204b). Therefore, Many researchers leverage the advantages of graph structures in handling irregular and non-uniform data, converting scene features into graph-structured data to accurately represent the interaction relationships between vehicles (Ha et al. 2025; Klimke, Völz, and Buchholz 2023). Such scene graphs are constructed with each vehicle as a graph node and the interaction influence relationships between vehicles as graph edges. The interactions are explicitly represented based on

the correlation of node features or fundamental communication relationships (Q. Liu et al. 2022; D. Zhou et al. 2022). Chen et al. (2021) relied on the perception and communication capabilities of autonomous vehicles to construct a CAV network as an interaction graph containing both local and global information. To further improve the information utilisation efficiency in complex interaction graphs, Xu et al. (Xu et al. 2023) integrated the logical relationships of vehicle intentions and actions into the perception-communication graph. The richer interaction information allowed the DRL model to demonstrate higher cooperative decision-making efficiency and robustness. To address the limitation of these methods relying on predefined graph structures, some researchers have introduced attention mechanisms to implicitly learn the correlations between node features, enabling the dynamic interaction process to be captured through adaptive weight adjustments (Liu et al. 2024a, 2024b; Zhang et al. 2024). However, these studies based on vehicle-vehicle interactions neglected the influence of road attributes on vehicle decision-making, especially in intersection environments, it is challenging to effectively distinguish their priorities, which can easily cause vehicle conflicts and deadlocks within intersections.

In this research, we integrate intersection geometric topology with dynamic graph structures to optimise interaction relationships among autonomous vehicles, thereby enhancing interaction efficiency within the system.

2.3. Safe decision-making in reinforcement learning

Traditional reinforcement learning lacks a practical method to specify which behaviours are allowed or prohibited. In most cases, the reward function is manually designed based on human prior knowledge to guide agent behaviours. However, under such conditions, risky decision-making is difficult to completely avoid, which limits the application of reinforcement learning in autonomous driving (Liu et al. 2024a, 2024b). To compensate for the deficiencies in reward function design, researchers often add a safety layer during action execution. Zhu et al. (2020) used kinematics-based safety distances as risk thresholds, forcing actions that violate the thresholds to be corrected to maximum braking acceleration to ensure the safety of action execution. Nguyen et al. (2023) adopted a finite state machine to map dangerous states to safe actions, using relative distance as the activation threshold to output more action patterns, which helps mitigate other risks brought by blind deceleration. However, forced corrections alter the state-to-action mapping relationship in the original model, which cannot be learned by the model, potentially disrupting the cooperation mechanism among agents in a multi-agent system. A more reasonable approach is to model the decision-making process as a Constrained Markov Decision Process (CMDP), which limits the learning process through safety constraints to avoid unsafe situations (Gu et al. 2024; Wang, Wang, and CHENG 2023). By modifying the learning objective, multi-objective optimisation problems can be converted into single-objective constrained problems (Chow et al. 2018; Y. Liu et al. 2024; Liu et al. 2023), or the policy update process can be constrained through trust region thresholds (Schulman 2015; Zhao et al. 2023). In this study, we incorporate a system efficiency objective that accounts for multi-agent safety risks based on interaction relationships. By enforcing a learning process that satisfies Lagrangian safety constraints, the AIM framework achieves conflict minimisation and efficiency maximisation.

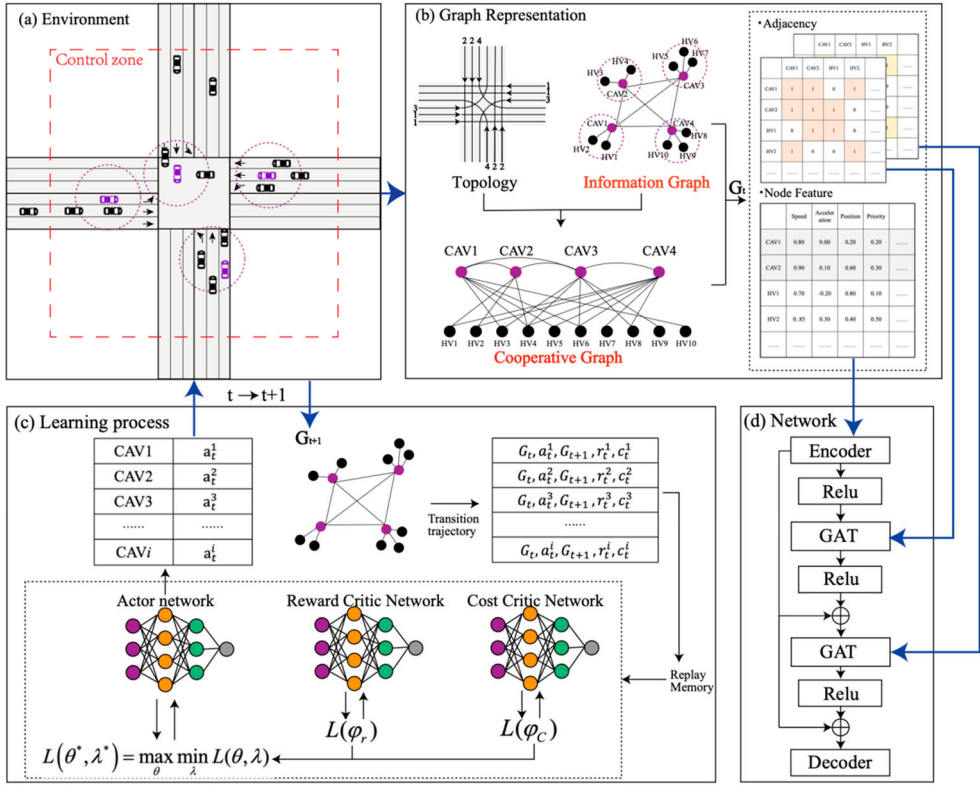


Figure 1. Cooperative driving framework in mixed traffic.

3. Methodology

3.1. Framework overview

To address the challenges of environment perception and safe decision making for autonomous vehicles at unsignalised intersection environments, this paper proposes a multi-agent cooperation control framework that includes a state representation module and a collaborative control module, as illustrated in Figure 1.

The state representation module transforms scene features into a graph-structured format through graph-based modelling including vehicle state features. The scene graph comprises both node and edge features, where node features represent the individual state information of each vehicle, and edge features capture the interaction relationships among vehicles. These interaction relationships result from the fusion of the information graph and the intersection topology. The information graph is constructed based on the inherent perception and communication capabilities of autonomous vehicles, while the intersection topology determines the cooperative relationship between vehicles.

The collaborative control module is responsible for mapping the scene state to vehicle actions. As shown by the relationship between (b) and (d) in Figure 2, after encoding node features, the global vehicle information is extensively aggregated through the information graph. Furthermore, leveraging intersection topology, the graph attention network

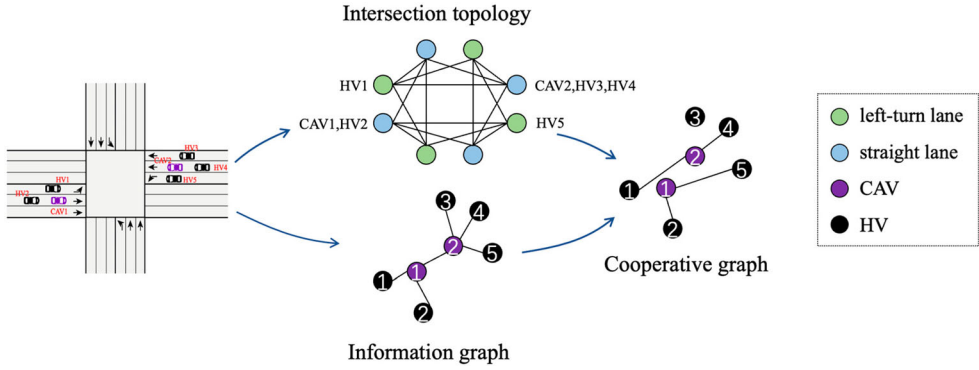


Figure 2. The construction process of the cooperative graph.

facilitates dynamic interactions and cooperation among vehicles. Finally, the decoder independently transforms the aggregated cooperative features of each node into the corresponding vehicle's action. This framework leverages global scene information and local cooperation relationships to achieve multi-vehicle cooperation during the approach process to intersections, thereby avoiding deadlock and collisions at unsignalised intersections through the control of CAV longitudinal behaviour.

The cooperation process needs to satisfy both systemic efficiency and safety benefits, but given the current development level of autonomous driving, safety requirements are usually more important for autonomous vehicles. We incorporate safety as a decision-making constraint and model the cooperative process as a Constrained Markov Decision Process (CMDP), facilitating the learning of safe behaviours through an additional cost critic network.

3.2. Hybrid graph attention representation

Effective cooperation among multiple agents in complex scenarios requires high-frequency and high-quality information exchange. GNN is an effective approach to address this problem. On one hand, GNNs can efficiently model non-Euclidean structures such as social networks, information networks, and traffic networks. In addition, GNNs perform extensive information propagation through the graph data structure $G = (V, E)$, where V is the set of nodes and E is the set of edges (J. Zhou et al. 2020). GNNs aggregate information from neighbouring nodes in various ways and update the node's own features. The propagation process of each GNN layer can be expressed as:

$$h_i^{(l+1)} = \text{UPDATE}(h_i^{(l)}, \text{AGGREGATE}(\{h_j^{(l)} | j \in \mathcal{N}(i)\})) \quad (1)$$

Here $h_i^{(l)}$ denotes the feature of node i ($i \in V$) at layer l . $\mathcal{N}(i)$ represents the neighbourhood of node i . AGGREGATE and UPDATE are functions defined by the specific GNN architecture.

Building on the above principles, previous researchers have developed various aggregation and update methods to adapt to the characteristics of different scenarios. The graph attention network (GAT) is particularly well-suited to traffic scenario due to the attention mechanism allows for dynamic adjustment of the importance weights of information

(Veličković et al. 2018). GAT assigns different attention weights to neighbouring nodes, enabling a weighted summation that achieves more effective aggregation.

$$h_i^{l+1} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{l+1} W h_j^l \right) \quad (2)$$

where σ is activate function, and the attention weight α_{ij} is dynamically adjusted based on the features of the target node and its neighbouring node. Specifically, it is computed as follows:

$$\alpha_{ij}^{l+1} = \frac{\exp(\text{LeakyReLU}(a^T [W h_i^l \parallel W h_j^l]))}{\sum_{k \in \mathcal{N}(i)} \exp(\text{LeakyReLU}(a^T [W h_i^l \parallel W h_k^l]))} \quad (3)$$

a is a learnable attention vector; w is a shared linear transformation matrix, $(\cdot \parallel \cdot)$ represents concatenation. Compared to the fixed weight design in methods such as GCN and GraphSAGE, this dynamic adjustment enables GAT to have a more efficient information transmission mechanism (Veličković et al. 2018). In traffic scenario, it can assist vehicles in identifying cooperation partners in highly complex and dynamic interaction environments.

In intersection environment, we define the relational features in the scene as a graph data structure. The nodes V represent CAVs or detectable HVs, and the node features h correspond to vehicle state information. The edges E represent the relationships between vehicle nodes. This relationship is based on the vehicles' perception of their surroundings, often through their communication and sensing capabilities (Q. Liu et al. 2022). Specifically, CAV i establishes an information exchange relationship e_{ij}^{com} with other CAV j through communication devices, and an information exchange relationship e_{ij}^{pre} with surrounding vehicle j through sensing devices. Based on these communication and perception information exchange relationships, an information graph e_{ij}^{info} integrating global scene information is established.

$$e_{ij}^{info} = e_{ij}^{com} \vee e_{ij}^{pre} \quad (4)$$

where the logical disjunction $\vee$ ensures that an information link is established if either communication or sensing connection exists.

For intersection areas where lane changes are restricted, the cooperation relationships within the information graph often include a significant amount of ineffective cooperation, which complicates the solution process. In general, intersections exhibit explicit lane connectivity, and the crossing relationships among lanes determine the conflict between vehicles. Therefore, the vehicle position information is mapped into conflict relationships e_{ij}^{con} through the topological connections of the lanes at the intersection. This mapping supports deeper modelling of information exchange relationships to determine whether cooperation between vehicles is required. The cooperation relation e_{ij}^{coo} is defined as:

$$e_{ij}^{coo} = e_{ij}^{info} \wedge e_{ij}^{con} \quad (5)$$

where the logical conjunction operator $\wedge$ ensures that cooperation is considered only when both conflict and information exchange conditions are satisfied between two vehicles. It is worth noting that, to avoid the negative impact of excessively large graph structures, when a single vehicle has more than 20 neighbours, only the 20 neighbours closest to

the intersection are retained. Figure 2 illustrates a simple example of cooperative graph generation.

As described above, the scene graph information is propagated and aggregated through a Graph Attention Network (GAT), enabling effective information exchange among vehicles. The dynamic weights of vehicle interaction relationships are computed according to Equation (6). As the initial step, a shared linear transformation is applied to each node i , parameterised by a trainable weight matrix $W \in \mathbb{R}^{f' \times f}$, where f and f' represent the dimensions of the original and transformed feature spaces, respectively.

$$h'_i = Wh_i \quad (6)$$

Based on features h' after linear transformations and attention mechanism $a : \mathbb{R}^{f'} * \mathbb{R}^{f'} \rightarrow \mathbb{R}$, attention coefficients are computed between each CAV and its adjacent CAVs. The cooperation relationship constraint e_{ij} is incorporated into the attention calculation to ensure that only relevant interactions are considered.

$$\varepsilon_{ij} = \begin{cases} \text{LeakyRelu}(a^T[h'_i \parallel h'_j]), & \text{if } e_{ij} = 1 \\ -\infty & \text{otherwise} \end{cases} \quad (7)$$

where leakyReLU is an activation function. For nodes with no collaborative relationships, the attention is set to $-\infty$, which ensures that the weight is 0 after softmax normalisation. To make coefficients easily comparable across different nodes, we normalise them across all neighbouring nodes $\mathcal{N}(i)$ using the softmax function:

$$a_{ij} = \text{softmax}(\varepsilon_{ij}) = \frac{\exp(\varepsilon_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(\varepsilon_{ik})} \quad (8)$$

As described in the aggregation process of Equation (2), the features of interacting CAVs are aggregated with dynamic weights to update the features of the target CAV. In this process, the multi-head attention mechanism is used to enhance the model's expressiveness. Each head independently fuses interaction features, and the outputs from n heads are averaged to obtain the final node feature representation.

$$h'_i{}' = \sigma \left(\frac{1}{n} \sum_{k=1}^n \sum_{j \in \mathcal{N}(i)} a_{ij}^k h_j'^k \right) \quad (9)$$

3.3. Lagrangian method

For the CMDP with safety constraints, a Lagrangian relaxation-based policy optimisation technique is employed to ensure the stability of the training process. A class of stationary policies π_θ parameterised by a parameter θ is considered. For the given π_θ , the objective function is defined via the infinite horizon discounted reward criterion:

$$J^r(\pi_\theta) = E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) | s_0 \sim \mu, a_t \sim \pi_\theta, \forall t \right] \quad (10)$$

Similarly, the objective function of the cost function is defined as follows:

$$J^c(\pi_\theta) = E \left[\sum_{t=0}^{\infty} \gamma^t C(s_t, a_t, s_{t+1}) | s_0 \sim \mu, a_t \sim \pi_\theta, \forall t \right] \quad (11)$$

Then $J^r(\pi_\theta), J^c(\pi_\theta) \in \mathbb{R}$. Let $d > 0$ denote a prescribed threshold below we want $J^r(\pi_\theta)$ to abide by. The constrained optimisation problem then is the following.

$$\max_{\theta} J^r(\pi_\theta) \text{ s.t. } J^c(\pi_\theta) \leq d \quad (12)$$

the associated Lagrangian can be defined as:

$$L(\theta, \lambda) = J^r(\pi_\theta) - \beta(J^c(\pi_\theta) - d) \quad (13)$$

where $\beta \in \mathbb{R}^+$ is the Lagrange multiplier. In terms of the Lagrangian, the goal is to find a tuple (θ^*, β^*) of the policy and Lagrange parameter such that

$$L(\theta^*, \beta^*) = \max_{\theta} \min_{\beta} L(\theta, \beta) \quad (14)$$

$J^r(\pi_\theta)$ and $J^c(\pi_\theta)$ as specified in (10) and (11) are not a priori knowledge, and need to be estimated by a neural network. As described by (Jayant and Bhatnagar 2022), It is challenging to find a globally optimal sadd point when θ serves as the parameter of a neural network. Therefore, the objective is to identify a locally optimal saddle point that satisfies Equation (14) a local neighbourhood $H_{\varepsilon_1, \varepsilon_2}$ which is defined as follows:

$$H_{\varepsilon_1, \varepsilon_2} \triangleq \{(\theta, \lambda) | \theta - \theta^* \leq \varepsilon_1, \beta - \beta^* \leq \varepsilon_2\} \quad (15)$$

where $\varepsilon_1, \varepsilon_2 > 0$, a local (θ^*, β^*) tuple can be found during gradient search.

3.4. DRL scheme and formulation

At time step t , each vehicle i in the multi-agent system independently interacts with the environment to obtain the vehicle state s_t^i and its relationships. By sharing information between agents, a scene graph $G_t = (S_t, E_t)$ is constructed, where $S_t = [s_t^i]_{i=1}^n$, $E_t = [e_t^{ij}]_{i,j=1}^n$. Through the transmission and fusion of state information, the new node features corresponding to agent i are used to generate the action a_t^i , which yields the updated vehicle state s_t^i , along with the individual reward r_t^i and cost c_t^i . The collection of individual state transition trajectories $[s_t^i, E_t, a_t^i, s_{t+1}^i, E_{t+1}, r_t^i, c_t^i]$ forms a scene state transition trajectory $[G_t, a_t, G_{t+1}, r_t, c_t]$, which is utilised for the training process.

Multi-agent learning system adopts Actor-Critic architecture to realise the paradigm of centralised training and decentralised execution. Actor-Critic architecture consists of an actor network with parameter θ and a critic network with parameter φ . The actor network generates an action a_t or the probability distribution of actions $\pi_\theta(a_t | s_t)$ based on scene graph. The critic network is used to evaluate multiple actions provided by the actor network, but outputs an estimated state value $V_\varphi(s_t)$ of scene, this aligns with the characteristics of centralised training and helps vehicles explore actions that are consistent with the overall system's objectives. The proximal policy optimisation (PPO) algorithm was used to train the DRL agent. The safe DRL framework involves two critic networks, which are used to estimate reward-based value function $V_{\varphi_r}^R$ and cost-based value function $V_{\varphi_c}^C$. The L_1 loss is adopted

to estimate value functions as follows:

$$L(\varphi_r) = \frac{1}{2} \left(\sum_{t=i}^{i+k-1} \gamma^{t-i} R_t + \gamma^k V_{\varphi_r}^R(s_{t+1}) - V_{\varphi_r}^R(s_t) \right)^2 \quad (16)$$

$$L(\varphi_c) = \frac{1}{2} \left(\sum_{t=i}^{i+k-1} \gamma^{t-i} C_t + \gamma^k V_{\varphi_c}^C(s_{t+1}) - V_{\varphi_c}^C(s_t) \right)^2 \quad (17)$$

where γ is the discount factor. The actor network is updated by maximising the objective function $J^r(\pi_\theta), J^c(\pi_\theta)$ which are clipped by PPO:

$$J^r(\pi_\theta) = \hat{E}_t[\min(r_t(\theta)\hat{A}_t^R(s_t, a_t), \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon)\hat{A}_t^R(s_t, a_t))] \quad (18)$$

$$J^c(\pi_\theta) = \hat{E}_t[r_t(\theta)\hat{A}_t^C(s_t, a_t)] \quad (19)$$

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (20)$$

where $r_t(\theta)$ is the probability ratio of the current policy $\pi_\theta(a|s)$ and old policy $\pi_{\theta_{old}}(a_t|s_t)$. The function $\text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon)$ modifies the surrogate objective by clipping the probability ratio to minimise the gap between the old and new policies, reducing the influence of large gradient on update stability. ε is the clipping parameter. $\hat{A}_t$ is the advantage function of policy at time t :

$$\hat{A}_t^R(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^R, \delta_t^R = r_t + \gamma V_{\varphi_r}^R(s_{t+1}) - V_{\varphi_r}^R(s_t) \quad (21)$$

$$\hat{A}_t^C(s_t, a_t) = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^C, \delta_t^C = c_t + \gamma V_{\varphi_c}^C(s_{t+1}) - V_{\varphi_c}^C(s_t) \quad (22)$$

3.4.1. State space

The state of the scene represented by the graph data is the input of the model. The state of the scene $G_t = (S_t, E_t)$ include node feature S_t and adjacency matrix E_t . Node feature S_t is the set of vehicle status information s_i^t in the scene, s_i^t includes speed, acceleration, position, lane, the priority of the corresponding lane, and the type of vehicle, denoted as $s_i^t = (v_i^t, a_i^t, d_i^t, l_i^t, p_i^t, type_i)$. To improve the stability of the training process, each element is scaled to keep its value within the range $(-1, 1)$. It is determined as follows:

- (1) $v_i^t = \frac{\tilde{v}_i^t}{v_{max}}$ is the normalised speed for each vehicle, where $\tilde{v}_i^t$ is the real speed of vehicle i at time t . v_{max} is the max speed of vehicle.
- (2) $a_i^t = \frac{\tilde{a}_i^t}{a_{max}}$ is the normalised acceleration for each vehicle, where $\tilde{a}_i^t$ is the real speed of vehicle i at time t . a_{max} is the max speed of vehicle.
- (3) $d_i^t = \frac{\tilde{d}_i^t}{d_{con} + d_{jun}}$ is the relative position information of the vehicle within the entry control range, where $\tilde{d}_i^t$ is the distance travelled by the vehicle i in the control area at t , d_{con} is the control area of entrance lane, d_{jun} is the range of junction.
- (4) l_i^t is lane information, this is a discrete variable. The four entrances of the intersection and the lanes of each entrance are coded separately by 'one-hot' method, denoted

as $I_i^t = [0, 0, 0, 0, 0, 0, 0]$. The first four digits of the vector are used to represent the entrance to the intersection, and the last three digits are used to represent the lane. The entrance and lane occupied by the vehicle are coded as 1, for example, when the vehicle approaches an intersection in the left-hand lane of the left arm, they are represented as $[0, 0, 0, 1, 0, 0, 1]$. This method can effectively eliminate the sequential relationship between the lanes.

- (5) $p_i^t = \frac{\tilde{p}_i^t}{p_{max}}$ is the relative priority of the lane in which the vehicle is currently lated. $\tilde{p}_i^t$ is the real priority of the lane, p_{max} is the max priority value of lane in intersection.
- (6) $type_i$ is the type of vehicle, CAV is 1 and HV is 0.

Adjacency matrix E_t is used to represent the connections between graph nodes in the scene graph, and it indicates whether there is interaction and cooperation between the corresponding CAVs. It is an $N \times N$ binary matrix, matching the dynamic number of vehicles in the scene, where an interaction between vehicles is represented by 1, and no interaction is represented by 0. To enhance the efficiency of interaction and cooperation in complex scenarios, this paper employs an adjacency construction method based on road topology relationships, which is detailed in Section of Hybrid Graph Attention Representation.

3.4.2. Action space

The execution action a for CAVs is acceleration, which is a continuous action space that adheres to the maximum acceleration constraints of the CAVs, $a \in [a_{min}, a_{max}]$. The model outputs a set of actions for all individual CAVs, but each action component in the set is generated independently of the other actions, relying on each node state after information propagation.

3.4.3. Reward function

The reward function plays a critical role in the learning process, guiding the agent to adopt cooperative behaviours that align with the efficiency and safety requirements of intersection management.

A higher efficiency reward is assigned when the vehicle travels closer to the maximum allowed speed, and is defined as follows:

$$r_v^i = \begin{cases} \frac{v_i - v_{min}}{v_{max} - v_{min}}, & v_{min} \leq v \leq v_{max} \\ -1, & v < v_{min} \text{ OR } v_{max} < v \end{cases} \quad (23)$$

where v_i is the current speed of vehicle i , v_{max} and v_{min} are the maximum and minimum speed limit of vehicle respectively. Vehicles with speeds below v_{min} are penalised to discourage them from reducing speed or stopping solely to gain safety advantages, which may lead to overly conservative or lazy behaviour.

CAVs face two types of conflicts at unsignalised intersections: crossing conflicts and following conflicts. Crossing conflicts arise from interactions between vehicles on different lanes, typically described using the Post Encroachment Time (PET) metric. PET represents the time between when a vehicle leaves the conflict area or point and when a following vehicle reaches the same area or point. Following conflicts arise from interactions between vehicles on the same lane, commonly described using the Time to Collision (TTC) metric. TTC represents the remaining time before a collision occurs if the vehicles maintain their

current speed and direction. Therefore, the risk cost includes PET cost and TTC cost, combined for vehicle i and its interacting vehicles. TTC typically applies to the follower in a following relationship, so only the follower will be penalised. TTC cost is denoted as:

$$c_T^i = 1 - \frac{TTC_i}{TTC_{lim}} \quad (24)$$

Unlike TTC, the same PET value applies to both vehicles in a conflict pair, however, assigning the same penalty to both vehicles is not ideal for the learning process. Therefore, based on the priority relationships of the road topology, PET cost is applied to vehicles with lower priority.

$$c_p^{ij} = \begin{cases} PET_{lim} - PET_{ij}, & \text{if } PRI_{ij} < 0 \text{ and } PET_{lim} > PET_{ij} \geq 0 \\ 1, & \text{if } PRI_{ij} < 0 \text{ and } -PET_{lim} \leq PET_{ij} < 0 \\ 0, & \text{otherwise} \end{cases} \quad (25)$$

where $PET_{ij} \geq 0$ indicates that vehicle i arrives at the conflict zone earlier than vehicle j , while $PET_{ij} < 0$ implies a later arrival. PRI_{ij} represents vehicle i relative to vehicle j , with a negative value indicating lower priority and a positive value indicating higher priority. PET_{lim} is the PET penalty threshold. Within this threshold range, the penalty function guides the vehicle to follow the priority order while gradually reducing the level of conflict. For the vehicles beyond this threshold, the conflict risk is relatively low, and enforcing strict priority may lead to excessive efficiency loss. We define the maximum value of c_p^{ij} among the neighbouring nodes $\mathcal{N}(i)$ as the final PET cost of vehicle i .

$$c_p^i = \max_{j \in \mathcal{N}(i)} c_p^{ij} \quad (26)$$

Accident cost c_A is used to prevent collisions from occurring, assigning a large penalty value of 10. It also serves as a settlement reward in the reward function. When a collision occurs, the training process ends the current episode and moves to the next one early. Like PET, both vehicles involved in the collision are penalised, with penalties applied only to vehicles with lower priority.

To enhance safety in the learning and policy updating processes, we incorporate safety constraints into the policy update process, treating the original problem as a constrained optimisation issue, as shown in Equation (12). The safety cost function is defined as follows:

$$C = \omega_T c_T + \omega_P c_P + \omega_A c_A \quad (27)$$

In this study, we define the reward including efficiency reward r_v and safety reward r_c as:

$$R = \omega_v r_v - \omega_c C \quad (28)$$

4. Simulation and discussion

4.1. Experiment design

The simulation experiment was carried out in the traffic simulator SUMO. The intersection unit in this experiment is the unsignalised intersection with four approaches, as shown in Figure 3. The length of each approach is 500 metres and includes two straight lanes and

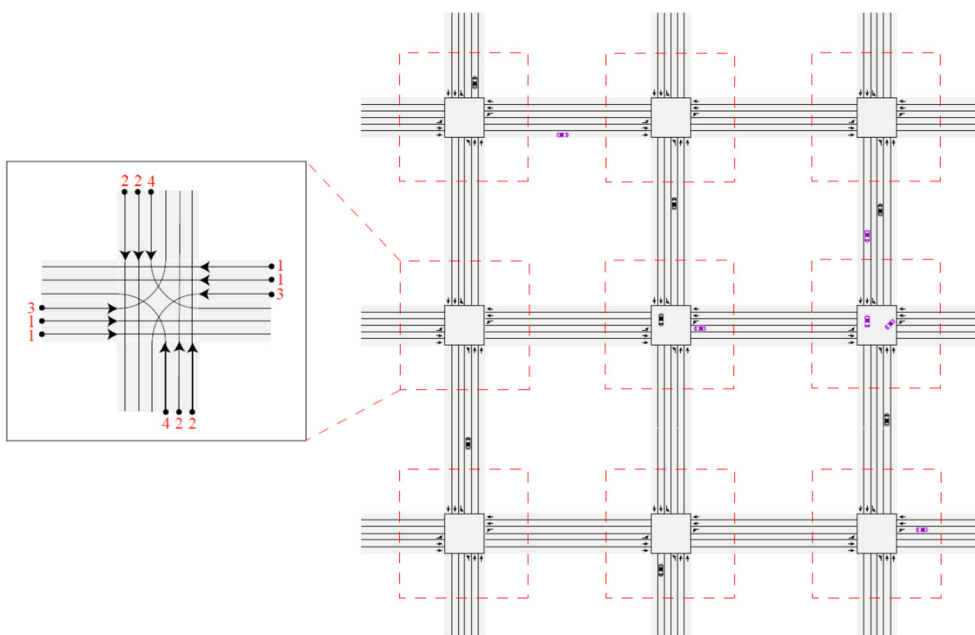


Figure 3. Traffic environment and Intersection topology.

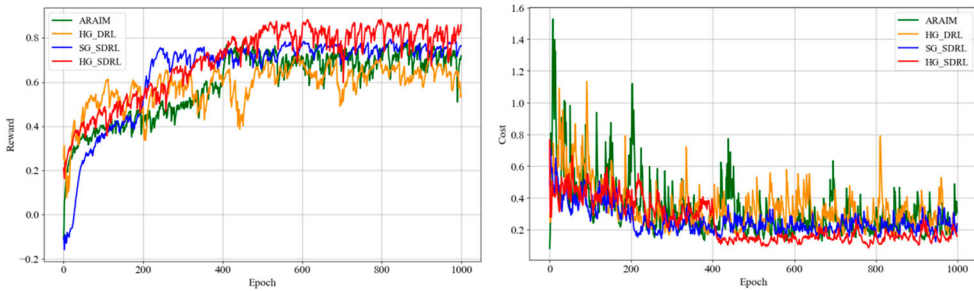
one left-turn lane. The lane priority is set as illustrated, with 1 representing the highest priority and 4 representing the lowest priority. The control area extends 150 metres from the intersection, within which lane changes are not permitted.

In this study, four baseline models were implemented for comparison with the proposed method. The Conflict Point-based First-Come-First-Served (CP-FCFS) model represents a traditional approach, being an improved version of FCFS based on lane conflict relationships (Zhong, Nejad, and Lee 2021). This model considers the order of vehicles only within conflicting lanes, rather than globally, allowing vehicles in non-conflicting lanes to arrive simultaneously. An advanced AIM method (ARAIM) proposed by Guillen-Perez and Cano (2022c) as a baseline model for a multi-agent approach, but we limit the number of captured conflicts in the method to align with the proposed method. In addition, two other baseline models are simplified versions of the proposed method as ablation experiments to validate the performance of the proposed method. Different from the proposed model, HG-DRL adopts a conventional reinforcement learning strategy. HG-DRL is a reinforcement learning strategy based on a single graph. SG-SDRL is a safety-constrained reinforcement learning strategy based on a single graph, which is an interaction relationship graph constructed from vehicle communication and sensing, without relying on road topology.

The main parameters of the experiments are presented in Table 1. The parameter settings are largely based on empirical values from previous reinforcement learning studies. The choice of the cost limitation is informed by preliminary training conducted before the formal experiments. During training, the cost constraint decays from an initial value of 1–0. The initial value is based on the cost function. It was observed that values smaller than 0.2 did not further improve the experimental results. The PET limitation was determined through comparative experiments, which are presented in the following sections,

Table 1. Main parameters of the experiments.

Parameters		Value
Environment	Traffic inflow rate	0.5 veh/s
	Speed range	[2 m/s, 16 m/s]
	Acceleration range	$[-3 \text{ m/s}^2, 3 \text{ m/s}^2]$
	minimum gap of IDM	2.5 m
	minimum headway time of IDM	1 s
DRL	Discounter factor γ	0.97
	λ -return parameter	0.95
	Actor learning rate	1e-4
	Critic learning rate	1e-4
	Clipping parameter ε .	0.2
	Hide layer units	512
	Number of attention head	4
	Optimiser	Adam
	Batch size	512
	Cost limitation d	0.2
	PET limitation	2
	Reward weight ω_V, ω_C	1
	Cost weight $\omega_T, \omega_P, \omega_A$	1

**Figure 4.** Reward and cost curves.

as it provides a reasonable balance between efficiency and safety. Since both the reward and cost functions are properly normalised, their weights are set to 1.

4.2. Performance comparison

4.2.1. Model training

Considering a decentralised framework, our model was trained exclusively on a single intersection scenario to ensure training efficiency. The training process was conducted in Ubuntu 20.04 with Python 3.9 and took approximately 24 h on a single NVIDIA GeForce RTX 3070 GPU. Figure 4 shows the reward and cost the reinforcement learning models during the training process. Compared to SG-SDRL, the proposed model demonstrates more stable and superior performance during training, benefiting from the guidance of priority. In comparison with ARAIM and HG-DRL, the proposed model achieves more stable and lower cost values throughout the training process, indicating that the additional cost constraints enable the model to learn safer cooperative strategies.

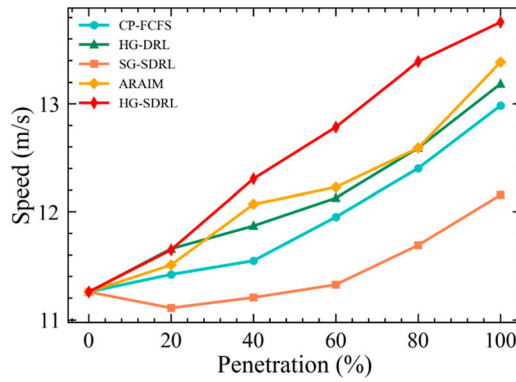


Figure 5. Average travel speed under different ROP.

Table 2. Statistics of performance metrics.

		CP-FCFS	ARAIM	HG-DRL	SG-SDRL	HG-SDRL
Speed	20%	11.42 ± 1.82 ($p = 0.008^*$)	11.51 ± 1.71 ($p = 0.034^*$)	11.66 ± 1.85 ($p = 0.032^*$)	11.11 ± 1.67 ($p = 0.000^*$)	11.65 ± 1.59
	60%	11.95 ± 1.12 ($p = 0.013^*$)	12.23 ± 1.10 ($p = 0.028^*$)	12.12 ± 1.22 ($p = 0.018^*$)	11.33 ± 1.56 ($p = 0.001^*$)	12.79 ± 1.01
	100%	12.98 ± 0.76 ($p = 0.001^*$)	13.38 ± 0.64 ($p = 0.010^*$)	13.19 ± 1.01 ($p = 0.000^*$)	12.16 ± 1.21 ($p = 0.001^*$)	13.76 ± 0.56
AveTTC	20%	2.35 ± 0.71 ($p = 0.018^*$)	2.37 ± 0.67 ($p = 0.013^*$)	2.36 ± 0.78 ($p = 0.030^*$)	2.41 ± 0.56 ($p = 0.014^*$)	2.46 ± 0.55
	60%	2.16 ± 0.79 ($p = 0.016^*$)	2.17 ± 0.68 ($p = 0.001^*$)	2.12 ± 0.80 ($p = 0.006^*$)	2.23 ± 0.61 ($p = 0.008^*$)	2.31 ± 0.59
	100%	2.06 ± 0.49 ($p = 0.001^*$)	2.02 ± 0.56 ($p = 0.000^*$)	1.95 ± 0.59 ($p = 0.000^*$)	2.09 ± 0.42 ($p = 0.003^*$)	2.10 ± 0.38
AvePET	20%	4.04 ± 0.89 ($p = 0.042^*$)	4.03 ± 0.80 ($p = 0.028^*$)	3.94 ± 1.02 ($p = 0.021^*$)	3.84 ± 0.79 ($p = 0.015^*$)	4.23 ± 0.83
	60%	3.60 ± 1.10 ($p = 0.012^*$)	3.43 ± 1.01 ($p = 0.002^*$)	2.87 ± 1.21 ($p = 0.000^*$)	2.99 ± 0.88 ($p = 0.000^*$)	3.83 ± 0.83
	100%	3.61 ± 0.67 ($p = 0.031^*$)	3.01 ± 0.80 ($p = 0.013^*$)	2.62 ± 0.82 ($p = 0.000^*$)	2.82 ± 0.76 ($p = 0.000^*$)	3.42 ± 0.76

mean ± std, $p^* < 0.05$.

4.2.2. System performance

To validate the performance of this method in mixed traffic flow, its efficiency and safety performance are compared and analysed against the baseline models under the same simulation environment. The results are averaged over 50 random experiments to ensure the reliability of the outcomes. We evaluate system efficiency using average speed, and assess safety performance using average PET and TTC. The results are summarised in Table 2. A paired T-test is conducted to verify that the experimental results of the proposed model differ significantly from those of the baseline models. The advantages of the proposed model are illustrated in Figures 5 and 6.

Figure 5 shows the system's average speed under different rates of penetration (ROP). The speed is positively correlated with ROP, which is consistent with the fundamental conclusions of many studies on autonomous driving in mixed traffic flows (Li, Cao, and Li 2021; S. Wang et al. 2022), as lower ROP are more influenced by HVs. Under HG-SDRL, the system exhibits the highest operational efficiency at any ROP. The lower average speed of

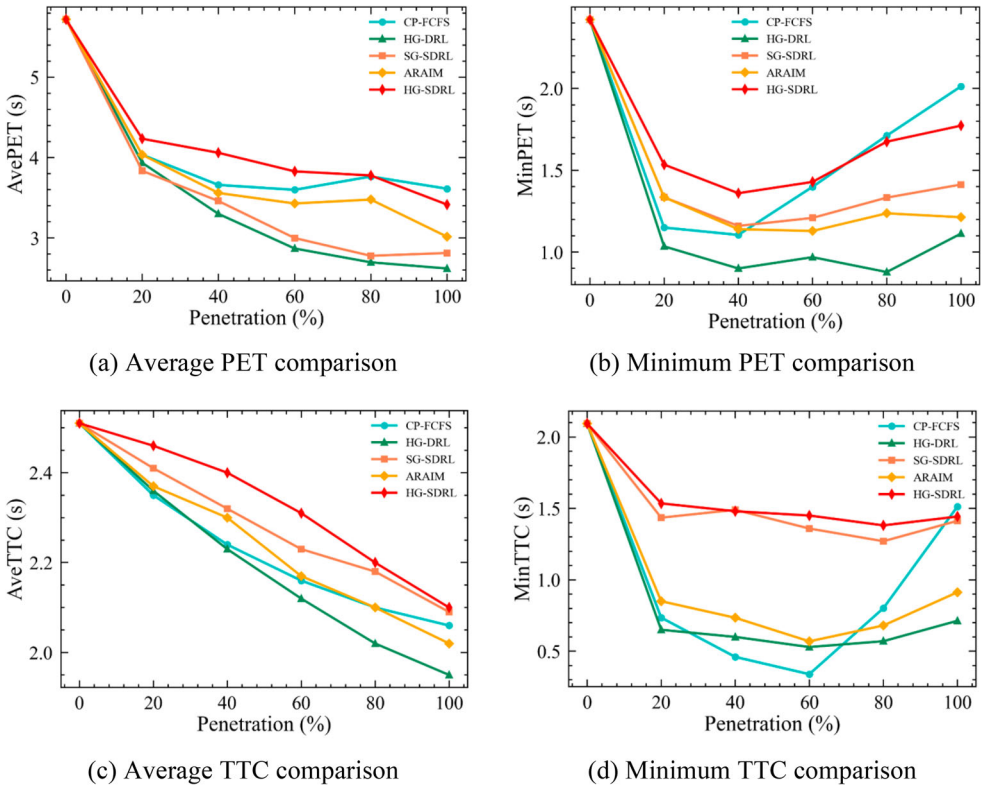


Figure 6. Safety performance under different ROP.

the CP-FCFS compared to HG-SDRL is primarily due to the strict adherence to the FCFS order by CAVs, where low-priority HVs in a mixed traffic environment adopt yielding strategies. CAVs following CP-FCFS may yield to low-priority HVs to maintain order, resulting in a significant loss of speed. Although ARAIM, HG-DRL and SG-SDRL also show an increasing trend in average speed with higher ROP, their overall efficiency is significantly lower compared to HG-SDRL, and even falls below the CP-FCFS strategy. The main reason is the lack of prior road topology information and safety constraint, which hampers effective guidance for scene representation and learning processes, making it difficult for agents to explore optimal strategies and leading to the ‘lazy agent’ problem. This manifests as CAVs adopting uniform strategies, such as continuous deceleration, in complex interaction scenarios, resulting in lower overall efficiency and even deadlock. More detailed reasons will be analysed in subsequent sections.

Due to public concerns about the safety performance of autonomous vehicles, safety is crucial for evaluating the quality of autonomous driving strategies. Figures 6(a,b) show the system’s average PET and minimum PET, respectively. The average PET is calculated based on PET values less than 5 s and describes the average gap between vehicles in the scene. The minimum PET reflects the severity of conflicts in the scene. As shown in Figure 6(a), the average PET decreases with increasing penetration rates. Based on the PET distribution in Table 2, under low penetration rates, a large number of HVs results in conservative and inefficient system interactions. As the proportion of autonomous vehicles

Table 3. The performance under different PET limitation.

Threshold (s)	1	2	3	5
Speed (m/s)	12.801	12.785	11.465	9.923
avePET (s)	2.181	2.312	2.545	2.431
minPET (s)	0.923	1.429	1.415	1.243

Under 60% penetration.

increases, they tend to adopt relatively aggressive driving strategies to pursue higher efficiency, which compresses the gap between vehicles at conflict points. This driving strategy is directly related to the PET threshold defined in Equation (25). Table 3 compares the model's performance under different PET thresholds. A smaller threshold encourages more aggressive behaviours, potentially leading to more severe conflicts. On the other hand, within the threshold range, vehicles are always required to follow the priority order, so a larger threshold results in greater efficiency loss.

More effective safety assessment is represented by the minimum PET shown in Figure 6(b), which represents the most dangerous situations encountered during interactions, smaller PET values imply a closer approach to collision. HG-SDRL and CP-FCFS perform worse at low penetration rates compared to high penetration rates due to the unpredictable behaviours of HVs. The excellent interaction capabilities of HG-SDRL allow it to maintain a higher minimum PET compared to CP-FCFS at low penetration rates. The stricter order constraints cause the CP-FCFS value to be slightly higher than HG-SDRL at 100% penetration rate, though the randomness of DRL cannot be avoided in any situation. Compared to other baselines, our AIM approach demonstrates a significant advantage in avoiding intersection conflicts, effectively validating the impact of incorporating priority guidance and safety constraints.

Figures 6(c,d) show the trends of average and minimum TTC with varying penetration rates. TTC values less than 3s are used for statistical calculations in the experiments (T. Liu et al. 2021). Average TTC is clearly negatively correlated with penetration rates, which is consistent with the trend of PET. HG-SDRL and SG-SDRL demonstrate relatively better performance because CAVs maintain a high level of attention to vehicles in the same lane to avoid conflicts during interactions. A detailed analysis of this attention will be provided in subsequent sections. The reasons for the lower TTC values of CP-FCFS and HG-DRL are different. FCFS relies on a fixed cooperation pattern that struggles to handle the unpredictable behaviours of HVs, a limitation that is absent at high penetration rates. Like the PET analysis, the defects of HG-DRL and ARAIM are due to the insufficient exploration under multiple objectives, which is not influenced by environmental penetration rates, as illustrated in Figure 6(d).

Additionally, we compared the model performance in a multi-intersection environment, as illustrated in Figure 3. Due to our decentralised framework and limited control scope, the proposed method can be easily extended to multi-intersection scenarios without requiring additional training. As shown in the Table 4, the model continues to exhibit significant advantages in the multi-intersection setting, consistent with its performance in the single-intersection case.

Through the analysis of the scene's average speed, PET, and TTC metrics, we have demonstrated that the designed multi-agent cooperative control strategy excels in

Table 4. Performance comparison in multi-intersection environment.

	Speed (m/s)	avePET (s)	minPET (s)	aveTTC (s)	minTTC (s)
CP-FCFS	11.658	3.489	1.245	2.33	0.566
ARAIM	12.354	3.506	1.056	2.20	0.762
HG-DRL	11.984	2.984	0.926	2.15	0.525
SG-SDRL	11.949	3.012	1.203	2.35	1.122
HG-SDRL	12.764	3.612	1.325	2.45	1.183

improving both the efficiency and safety performance of unsignalised intersections, showing higher robustness to changes in penetration rates in mixed traffic environments. This outstanding performance is attributed to the interaction and cooperation among CAVs. In the next section, we will attempt to describe the advantages of the proposed framework from a more microscopic perspective, based on the vehicle movement processes in the scene.

4.2.3. Process performance

We established an interaction scenario with 24 vehicles present under a 0.5 veh/s inflow rate for each entrance for the experiment. The experiment includes a pure CAV environment (Figure 7(a)) and a mixed traffic environment with a ROP of 0.5 (Figure 8(a)). In these figures, the purple box represents CAV, the yellow box represents HV, and the red number next to the vehicle represents speed. Additionally, we performed an extra analysis of the vehicle dynamic interaction processes in the pure CAV scenario, which benefits from the explicit representation of the interaction process provided by the attention mechanism.

Figure 7(a) shows the initial state of the pure CAV scenario. Figures 7(b)–(d) illustrate the detailed process under different strategies. The FCFS strategy follows a strict passage order and can be described as an inference process. Vehicles at the end of the inference chain are assigned too late, especially in scenarios where there are too many vehicles, as seen with vehicles 1 and 5 in Figure 7(b), leading to reduced passage efficiency. SG-SDRL and HG-SDRL exhibit noticeable differences in performance. Figure 7(c) shows that CAVs under SG-SDRL tend to adopt more conservative driving strategies overall. Left-turning vehicles, such as vehicle 12, enter the control area earlier but are assigned the latest passage times, which is the cause of the severe speed loss under the SG-SDRL strategy. In HG-SDRL, the introduction of road priority relationships allows more vehicles to follow these priority rules, as seen with vehicles 0 and 22 that entered earlier. But low-priority vehicles are not entirely constrained by priority, such as vehicles 6 and 12, pass through the intersection earlier than higher-priority vehicles in relatively safe conditions – conditions where there are no severe PET and TTC risks. We provide corresponding guidance thresholds in the reward function (22) (23). This appropriately aggressive driving style makes HG-SDRL more efficient than FCFS. Figure 7(e) shows the trend of the average speed across the scene throughout the process, with HG-SDRL demonstrating higher passage efficiency.

Figure 8(a) shows the vehicle distribution in the initial state of the mixed traffic scenario. Figures 8(b)–(d) illustrate detailed process under different strategies. In the mixed environment, HVs follow the road priority settings. The road priority conflicts with the FCFS principle. When vehicles arrive earlier, such as vehicle 6 in Figure 8(b), but the road priority is lower, they must give way to vehicles that arrive later, resulting in loss of efficiency. CAVs

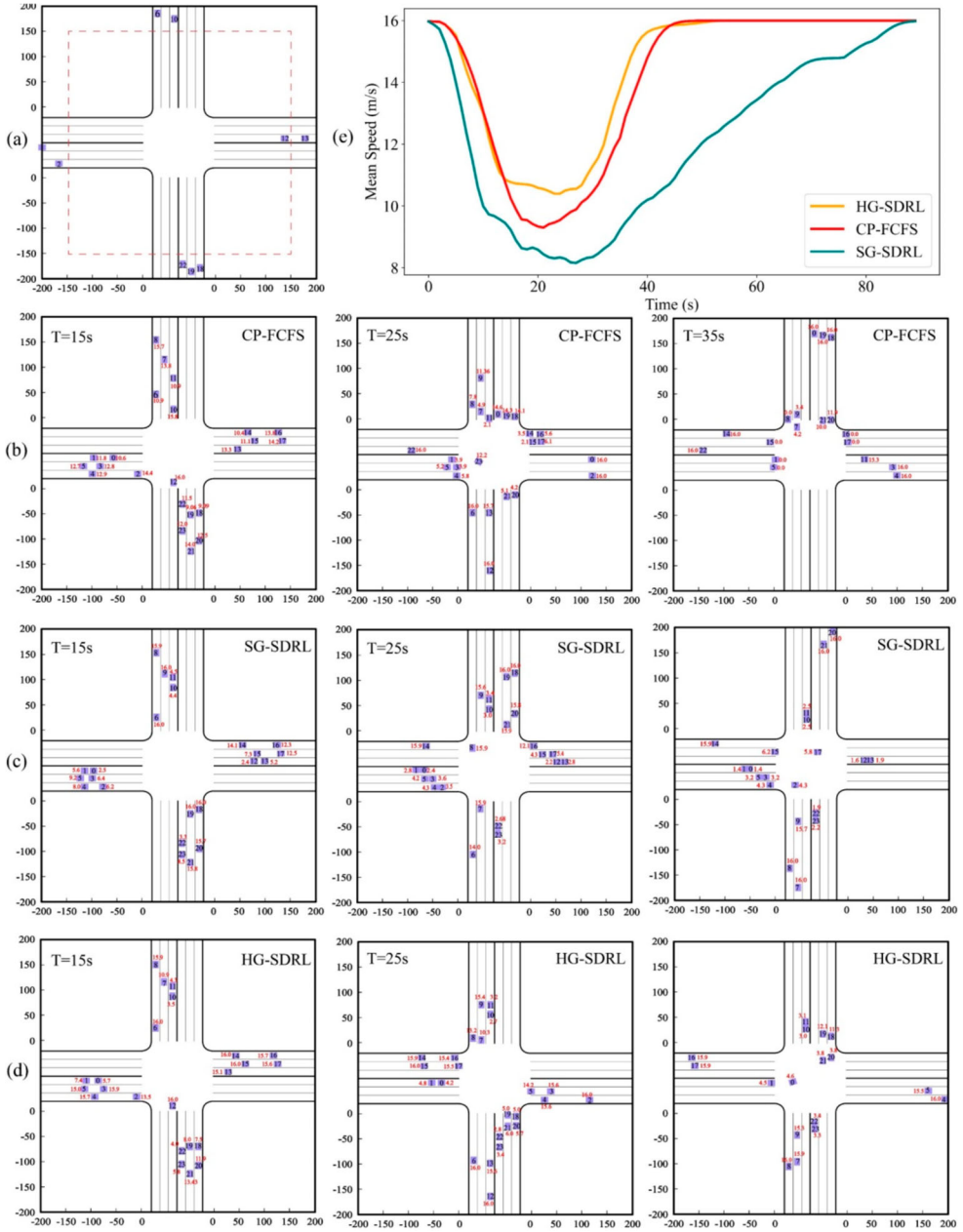


Figure 7. System process of different strategies under pure CAV environment.

under SG-SDRL, due to its conservative driving strategy, faces the same decision dilemma as FCFS when interacting with HVs.

In Figure 8(c), CAV2 and CAV3 get into a priority dilemma with HV 6 and HV 7, which causes CAV to slow down or stop and forms a long-term deadlock, as shown in the scene speed trend in Figure 8(e). On the other hand, the HG-SDRL strategy also integrates road priority relationships, aligning with the HDV strategy. The aggressive driving strategy of

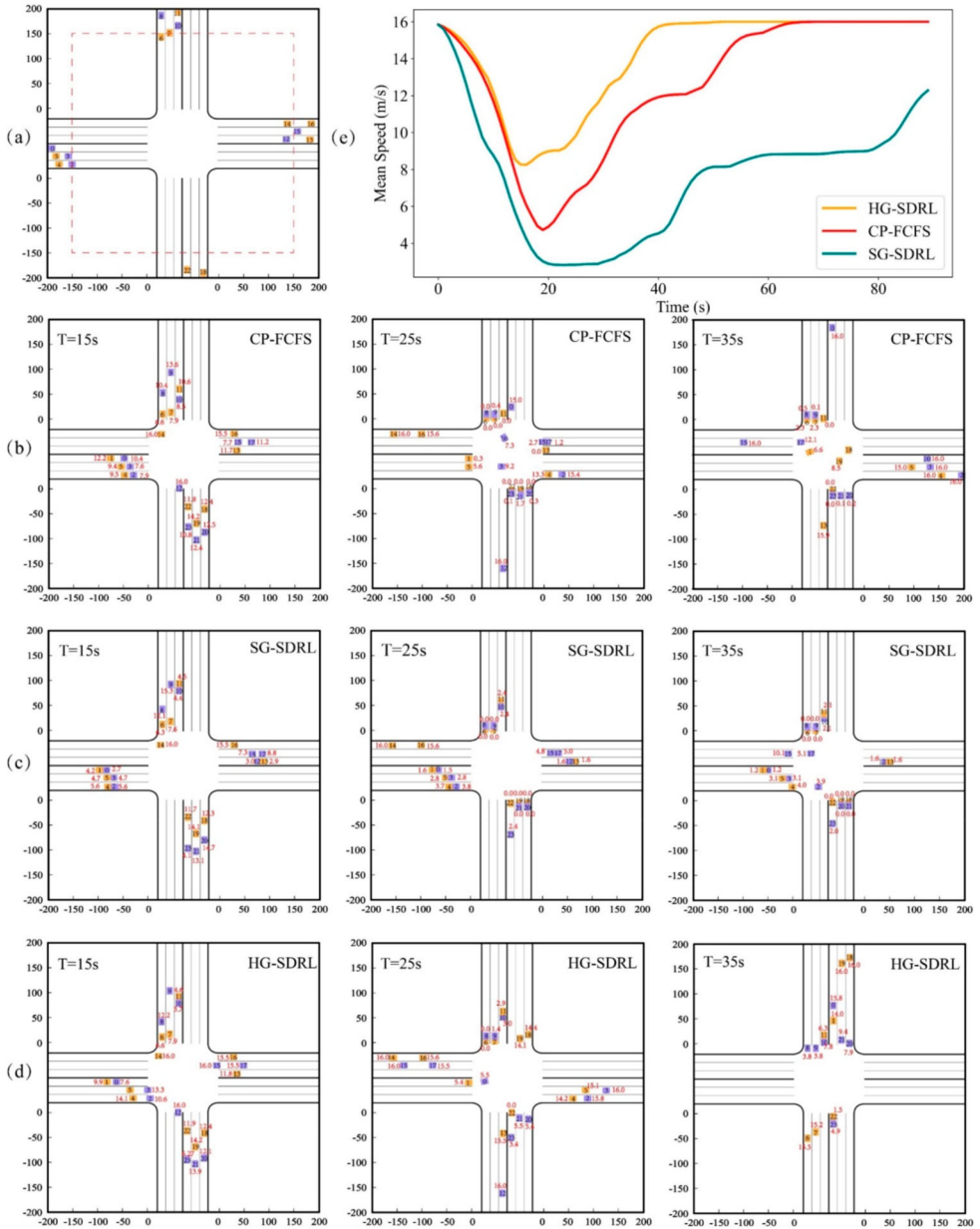


Figure 8. System process of different strategies under mixed environment.

CAVs effectively avoids deadlock situations with lower-priority HVs, allowing the HG-SDRL strategy to maintain high efficiency even in mixed traffic environments.

4.2.4. Attention analysis

The attention mechanism provides an explicit representation of the interaction process between vehicles. This explicit attention weight reflects which state information is more

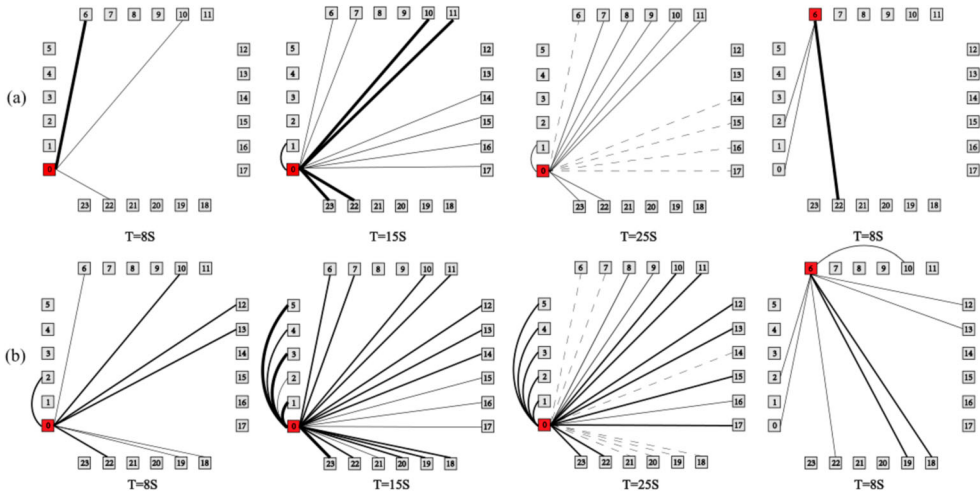


Figure 9. Attention changes during the system operation process.

effective during the learning process and helps to explain the formation of different driving styles.

We further analyse the processes shown in Figures 7(c) and (d) by focusing on vehicle 0 in the scenario. As shown in Figure 9(a) depicts the attention weight changes under HG-SDRL, and (b) shows the changes under SG-SDRL. The black lines represent interaction relationships, with the width of the black lines indicating the magnitude of the attention weights, a thicker black line denotes a stronger focus. The initial local subgraph for vehicle 0 includes vehicles 6, 10, and 22, with the highest attention weight assigned to vehicle 6, which has the closest arrival time. During the transition from state $T = 8s$ to $T = 15s$, vehicle 0, having a lower priority than vehicle 6, maintains a deceleration state, leading to a gradual reduction in the PET cost between vehicle 0 and vehicle 6. Simultaneously, due to changes in speed, vehicle 0 shifts its attention towards subsequent vehicles 10 and 11. With minimising potential conflicts through continuous interaction, vehicles reach a stable operational state, as shown in Figure 7, the state at $T = 25s$ illustrates the attention distribution of vehicles under stable conditions. Due to the reduced potential for conflict, vehicles maintain a balanced attention distribution towards all interacting entities.

In comparison, we find that greater attention weights tend to be assigned to vehicles with arrival times close to the intersection vehicle 6 in $T = 8s$ and vehicles 10, 11, and 23 in $T = 15s$ receive more focus from vehicle 0 due to their similar arrival times, making them potential conflict vehicles. Through gradually resolving these local conflict relationships during the evolution process, the final passing order is determined. When the potential conflict vehicles have similar arrival times, lower-priority interacting vehicles are assigned relatively higher attention weights. In the initial subgraph of vehicle 6, vehicle 22, as a potential conflict vehicle with a lower priority, receives higher attention. This is because high-priority vehicles are often unaffected by the behaviours of lower-priority vehicles and can largely maintain their current state of motion, while the movements of lower-priority vehicles are easily influenced by the behaviours of other vehicles, affecting the cooperation between interacting vehicles. By prioritising attention to vehicles with similar arrival

times at the intersection, the primary task of the vehicles becomes resolving conflict issues rather than adhering to other constraints like priority. The neglect of high-priority vehicles' attention also supports this, leading to more aggressive driving strategies for CAVs.

In the SG-SDRL model, although extensive communication and perception can help vehicles extract key interaction features from more scene information, excessive irrelevant information can interfere with inter-vehicle cooperation. Since there is no prior information to guide, Vehicles under SG-SDRL allocate attention weights based on the lane information and arrival time at the intersection. Vehicles on the same type of lane are assigned higher weights (e.g. left-turning vehicles focus more on other left-turning vehicles, and straight-going vehicles focus more on other straight-going vehicles). Additionally, vehicles on the same lane that arrive earlier at the intersection are given higher weights. The main reason might be that states with higher similarity are more likely to conflict. For example, in the initial state at $T = 8s$, the left-turning vehicle 0 assigns higher attention to vehicles on the left-turn lanes of each approach, but vehicle 6, as a potential conflict object, is not highly focused on, leading to the loss of key interaction features. Similarly, the straight-going vehicle 6 also ignores the potential conflict object 0. The high interaction between vehicles on the same lane results in distinctly different strategies for left-turning and straight-going vehicles. As shown in Figure 7(c), to ensure safety, left-turning vehicles yield to straight-going vehicles and are released last, regardless of whether they entered the control zone earlier. This leads to a significant decrease in intersection passing efficiency. In contrast to HG-SDRL, SG-SDRL's greater focus on lane information leads to more conservative driving strategies for CAVs.

The above analysis is sufficient to demonstrate that the introduction of road topology priors can help filter out more effective feature information from global characteristics. more effective feature clearly distinguishes the roles each vehicle undertakes in the cooperation process, thereby improving the issue of policy homogenisation in multi-agent systems, and enhancing the efficiency and safety of vehicle decision-making.

5. Conclusion

This paper proposes a multi-agent collaborative control framework that integrates graph representation learning and safe reinforcement learning to achieve automated management of unsignalised intersections in mixed traffic environments. This method considers the inherent uncertainty of human-driven vehicles through a graph-based scenario representation method and introduces additional road topology prior knowledge to help CAVs establish more stable cooperative relationships and learn potential driving pattern of human-driven vehicles, thereby improving system operational efficiency. The framework has high safety performance, primarily by integrating safety optimisation constraints within the CMDP and enhancing model safety performance through safe reinforcement learning training.

Compared with the FCFS strategy and other ablation models, the proposed method not only achieves more efficient dynamic collaboration with other CAVs but also demonstrates adaptability to the driving strategies of human-driven vehicles in mixed traffic flows. Experimental results under different ROP indicate that the proposed method exhibits superior performance in both system efficiency and safety. Moreover, stronger evidence from process experiments and attention analysis demonstrates that the proposed collaborative

control framework can address the dynamic cooperation of vehicles in complex environments, achieving automated management of unsignalised intersections in mixed traffic flows.

Compared to the traditional FCFS strategy and current advanced multi-agent methods, the proposed approach not only achieves more effective dynamic cooperation with other autonomous vehicles, but also shows strong adaptability to the driving behaviours of human-driven vehicles in mixed traffic flows. Experimental results under different ROP levels demonstrate that the method performs well in terms of both system efficiency and safety. Process experiments and attention analysis further indicate that the proposed cooperative control framework effectively addresses the challenge of dynamic coordination among vehicles in complex environments, enabling automated management of signal-free intersections in mixed traffic scenarios.

There are still several limitations in this study. The construction of interaction relationships relies on the assumption that autonomous vehicles possess perfect perception and communication capabilities, which is not feasible in current real-world environments. Moreover, the mixed intersection scenarios used in the study are idealised, while many complex and dynamic factors still exist, such as varying traffic densities, the diverse driving styles of human-driven vehicles, and different types of intersections. Investigating the scalability and adaptability of the proposed framework under these conditions represents a valuable direction for future research.

Disclosure statement

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